

Table 1. Flow rates and normal boiling point ranges for products from processing of 10,000 BPSD of Poseidon Crude (58,018 kg/h). Design NBP range and flow rates are from CHEMCAD.

Product	Composition	Spec NBP Range, °C	Design NBP Range, °C	Flow Rate, BPSD	Flow Rate, kg/h
LPG	C1 to C4	-162 to -1	-17 to -7	258	2,130
Naphtha	C5 to C12	30 to 200	80 to 130	2,773	29,317
Gas oils	C12 to C20	200 to 371	180 to 370	2,650	33,677
Residuum	C20 to C70	371 to 540	370 to 492	4,234	53,805
Sour Gas	H ₂ S, C1-C3, H ₂ O	---	-72 to 45	53	534
Sour Water	H ₂ O, H ₂ S	---	~100	31	450
Total	---	---	---	9,998	119,912

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Table 2. Sulfur and water content for distillate cuts from Poseidon Crude. Design values are from CHEMCAD.

Product	Sulfur, spec	Sulfur, design	H ₂ O, spec	H ₂ O, design
LPG	140 mg/kg [1]	186 mg/kg	10 mg/kg ^a [2] [3]	0.001 mg/kg
Naphtha	80 mg/kg [4]	172 mg/kg	0.005-0.05 vol% [4]	0.016 vol%
Gas oils	15 mg/kg [5]	55 mg/kg	0.01 vol% [5]	0.006 vol %
^a Must pass the Propane Dryness Test (cobalt bromide) or ASTM D 2713-70 (or 2713-91)				

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Table 3. Results of economic analysis of PYOIL process.

Capital Investment (millions of dollars)	
Purchased Equipment Cost	4.085
Working Capital, 15% of TCI	3.216
Total Indirect Costs	5.297
Total Direct Costs	12.582
Fixed Capital Investment	17.879
Total Capital Investment	21.095
Measures of Profitability	
Return on Investment, %	19.69
Payback Period, years	3.59
Net Present Worth, million \$, at 6% interest	23.336
Net Present Worth, million \$, at 12% interest	12.614
Internal Rate of Return, %	24.61
Profitability Benchmarks	
Bond Rate, after taxes, from Table 7-1, %	3.25
Preferred Stock Dividend, from Table 7-1, %	8.00
Common Stock Dividend, from Table 7-1, %	9.00
Minimum Acceptable Return, from Table 8-1, %	12.00
Ref. Payback Period, at MAR, years, eq 8-2c	4.14

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Table 4. List of purchased equipment costs from CHEMCAD for atmospheric crude and light ends units.

Number	Name	Description	CHEMCAD Unit	Cost
100	V-100	Desalter	Multipurpose Flash	\$534,516
101	V-101	Lower Column	TOWR Distillation Column	\$360,630
102	V-102	Upper Column	TOWR Distillation Column	\$290,844
103	V-103	Reflux Drum	Multipurpose Flash	\$60,036
104	V-104	Kerosene Stripper	TOWR Distillation Column	\$20,954
105	V-105	GO/D Stripper	TOWR Distillation Column	\$33,113
106	V-106	Lt Naphtha Recovery	TOWR Distillation Column	\$119,574
107	V-107	Lt Naphtha Stabilizer	TOWR Distillation Column	\$190,933
201	E-201	Lt Naphtha Cooldown	Heat Exchanger	\$4,345
204	E-204	Heavy Naphtha Cooldown	Heat Exchanger	\$7,230
205	E-205	Kerosene Cooldown	Heat Exchanger	\$8,142
207	E-207	GO/D Cooldown	Heat Exchanger	\$11,339
301	E-301	Lt Naphtha Energy Recovery	Heat Exchanger	\$29,827
302	E-302	Condenser Energy Recovery	Heat Exchanger	\$15,272
303	E-303	Upper Pump-around	Heat Exchanger	\$8,488
304	E-304	Heavy Naphtha Energy Recovery	Heat Exchanger	\$24,772
305	E-305	Kerosene Energy Recovery	Heat Exchanger	\$24,012
306	E-306	Lower Pump-around	Heat Exchanger	\$8,226
307	E-307	GO/D Energy Recovery	Heat Exchanger	\$111,498
308	H-308	Fired Heater	Fired Heater	\$1,136,449
309	E-309	Condenser	Air-Cooled Heat Exchanger	\$423,151
401	P-401	Crude Oil Feed	Pump	\$9,271
402	P-402	Desalted Crude	Pump	\$10,464
403	P-403	Lower Pump-around	Pump	\$8,244
404	P-404	Upper Pump-around	Pump	\$8,292
405	P-405	Reflux	Pump	\$8,627
406	P-406	Light Naphtha	Pump	\$9,045
407	P-407	VGO/RES	Pump	\$8,401
408	P-408	GO/D	Pump	\$9,400
409	P-409	Kerosene	Pump	\$8,449
410	P-410	Heavy Naphtha	Pump	\$9,103
411	P-411	Lt Naphtha	Pump	\$8,816
501	C-501	Lt Ends	Compressor	\$563,607

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Table 5. Utility usage rates from CHEMCAD for atmospheric crude and light ends units.

Number	Name	Description	Rate, \$/hr
308	H-308	Fired Heater	100.91
309	E-309	Condenser	90.94
401	P-401	Crude Oil Feed Pump	18.10
402	P-402	Desalted Crude Pump	0.28
403	P-403	Lower Pump-around Pump	0.78
404	P-404	Upper Pump-around Pump	0.04
405	P-405	Reflux Pump	0.06
406	P-406	Light Naphtha Pump	0.17
407	P-407	VGO/RES Pump	0.02
408	P-408	GO/D Pump	0.08
409	P-409	Kerosene Pump	0.01
410	P-410	Heavy Naphtha Pump	0.02
411	P-411	Lt Naphtha Pump	0.01
501	C-501	Compressor	18.10

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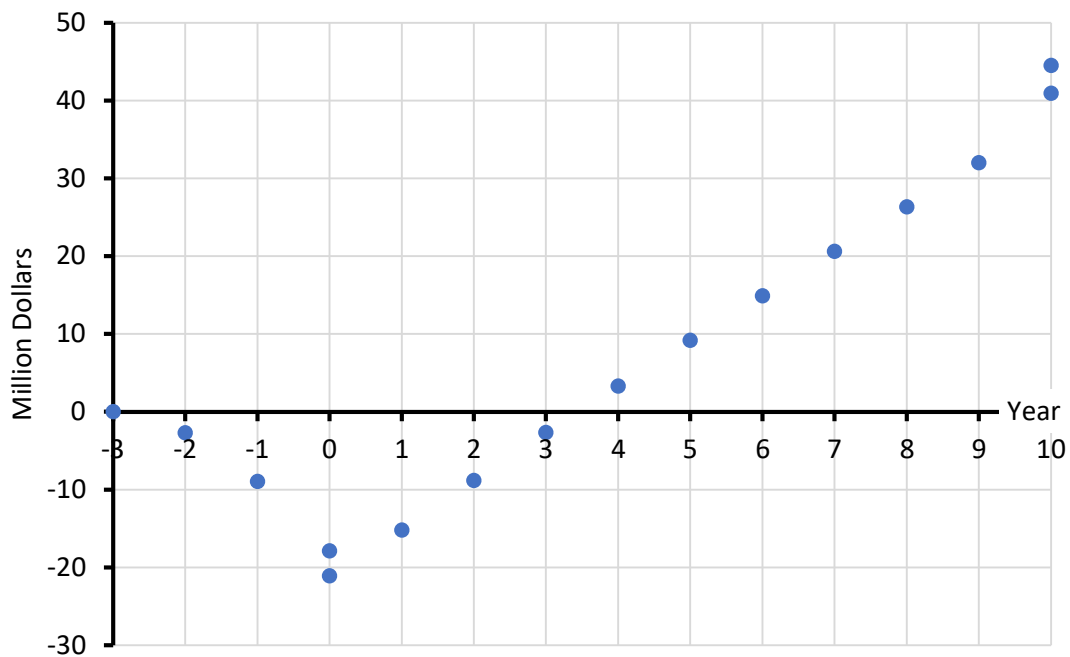


Figure 1. Cumulative cash position for atmospheric crude and light ends units processing 10,000 BPSD of Poseidon crude, as per Figure 6-2 in Peters, Timmerhaus, and West. The payback period (payout time) is 3.59 years.

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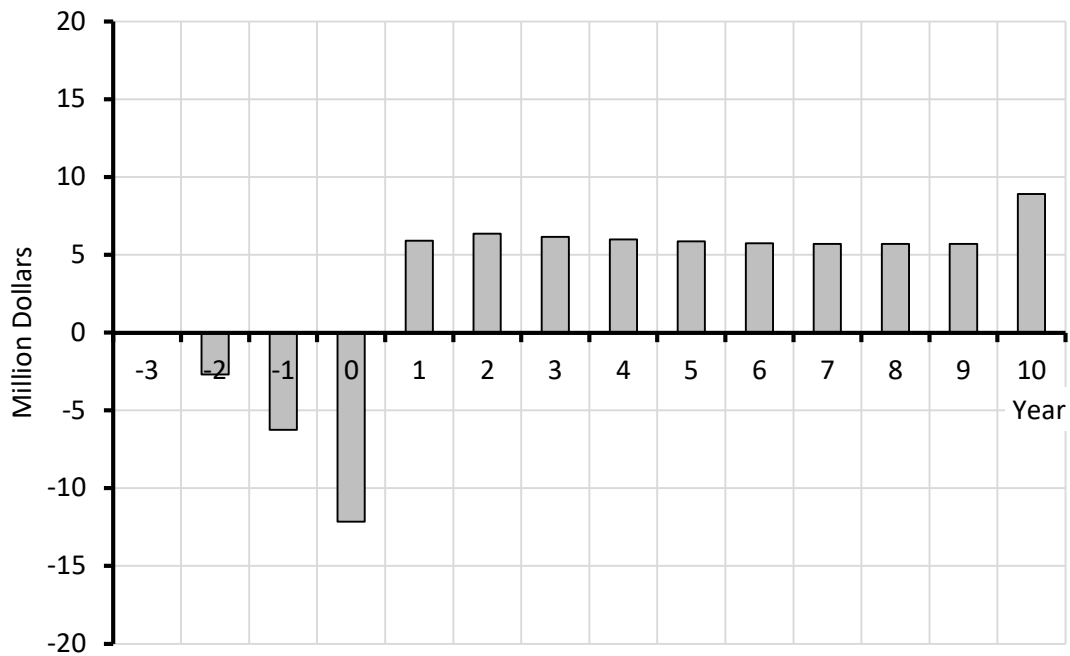


Figure 2. Annual cash flows (from colorful worksheet) for atmospheric crude and light ends units processing 10,000 BPSD of Poseidon crude, as per Figure 7-4 in Peters, Timmerhaus, and West.

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Table 6. Exposure and safety factors for crude oil and distillates.

Species or Fraction	Human Toxicity		Flammability Limits, % in air		Flashpoint, °C	Autoignition Temp., °C	NFPA 704 [6]		
	LD50	LC50	LFL	UFL					
Methane	N/A	500,000 ppm 2h mouse 325 g/m ³ 2h mouse [7] 658 mg/L 4h rat [8]	5.3 [9]	14.0 [9]	-188 [4]	537 [9]	2 [9]	4 [9]	3 [9]
Ethane	N/A	658 mg/L 4h rat [10]	3.0 [11]	12.5 [11]	-135 [11]	472 [11]	1 [11]	4 [11]	0 [11]
H ₂ S	N/A	713 ppm 1h rat [12] 673 ppm 1h mouse [12] 634 ppm 1h mouse [12] 444 ppm 4h rat 4h [12] 700 mg/m ³ 35 min rhesus monkey [13]	4.0 [13]	44.0 [13]	-82.4 [12]	260 [13] 232 [12]	4 [13]	4 [13]	0 [13]
LPG	N/A	> 30 mg/L 4h rat [14]	1.8 [15]	13.0 [15]	-5 [15]	400-450 [15]	2 [15]	4 [15]	0 [15]
Naphtha	>5,000 mg/kg rat (oral) [16] >2,000mg/kg rabbit (dermal) [16]	>5 mg/L 4h rat [16]	1.1 [17]	5.9 [17]	37.8 [16]	288 [17]	2 [16]	4 [17]	0 [16]
Gas Oil	>5,000 mg/kg rabbit (dermal) [18] >5,000 mg/kg rabbit (oral) [18]	>6 mg/L 4h rat inhalation [18]	0.6 [18]	6.5 [18]	52 [18] 52-96 [19]	257 [18] 257 [19]	1 [18]	2 [18]	0 [18]
Residuum	>2,000 mg/kg rabbit (dermal) [20] >5,000 mg/kg rat (oral) [20]	>94.4 mg/m ³ 5h rat [20]	N/A	N/A	204 [20] 158-275 [21]	485 [20]	0 [20]	3 [20]	0 [20]
Crude Oil	>2,000 mg/kg rabbit (dermal) [22] >4,300 mg/kg rat (oral) [23]	3.95 mg/L [24]	1 [25]	7 [25]	-29 [23] 0-40 [25]	454 [25]	2 [25]	3 [25]	0 [25]

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